

Deep reinforcement learning applied to statistical arbitrage investment strategy on cryptomarket

Gabriel Vergara, Werner Kristjanpoller*

Departamento de Industrias. Universidad Técnica Federico Santa María., Av. España 1680, Valparaíso, Chile

ARTICLE INFO

Keywords:

Statistical arbitrage
Pairs trading
Deep reinforcement learning
Cointegration
Investment strategies
Cryptocurrency market

ABSTRACT

In the complex and dynamic nature of financial markets where increasingly sophisticated investment strategies are required, deep reinforcement learning (DRL) has proven successful in generating real-time investment strategies, outperforming classical models. Alongside this, statistical arbitrage strategies exploit temporary market inefficiencies to generate returns. In this regard, a novel DRL-based arbitrage method has been developed.

This paper presents a unified framework that combines classical statistical arbitrage theory with DRL framework to generate investment strategies. The framework addresses the challenges of identifying similar asset portfolios, extracting signals indicating temporary price deviations, and determining optimal trading rules given market conditions. The proposed methodology involves constructing arbitrage portfolios based on cointegration relationships, removing signals from price series and portfolios, and using a DRL agent to make optimal decisions within a fixed time horizon.

The empirical analysis focuses on the cryptocurrency market, known for its volatility and risk. Results demonstrate that DRL agents can generate strategies with positive returns in out-of-sample periods 79.52% to 112.82% with no transaction cost, outperforming market benchmark Bitcoin 32.51% return, the best performing over the period Litecoin with 57.11% return and the worst performing Solana with a 35.70% loss. Moreover, these strategies effectively reduce risk, achieving higher risk-adjusted returns on individual assets. The strategies maintain positive returns when considering transaction costs, with the DRL agent outperforming the standard arbitrage strategy. The best-performing strategy is based on a Deep Q-Network (DQN) agent with a return of 18.39%, annualized volatility of 12.22%, and an annualized Sharpe ratio of 2.43. At the same time, Bitcoin holds an annualized volatility of 44.13% and a Sharpe ratio of 1.08.

Their randomness and coherence are studied to verify the robustness of the agents' decisions. The actions generated by the agents are not random but based on well-founded policies, which align with the obtained results.

1. Introduction

Gatev et al. (2006) [1] introduced the concept of pairs trading, a simple process that follows two guidelines. Firstly, two assets are sought whose prices have a joint movement throughout history in a formation period. Secondly, the price difference or spread between the assets is monitored during a trading period. If the prices diverge and the spread widens, the loser is bought, and the winner is sold. In the case of selected assets or stocks with an equilibrium relationship, the spread is expected to revert to its historical mean. Thus, positions are reversed, and profit can be obtained from this. Subsequently, a natural modification to pairs trading is to extend it to a set of assets rather than pairs, giving rise to statistical arbitrage. Statistical arbitrage encompasses a variety of investment strategies that identify and exploit temporary

price differences between similar assets using statistical tools, which is understood as profiting from market inefficiencies. While its most basic form corresponds to pairs trading, these strategies do not bet on the market direction but on temporary differences or misvaluations in assets during short periods of time.

Considering that every arbitrage strategy requires solving three fundamental problems: Given a large number of assets, which are the portfolios of similar assets that represent a joint movement? Given these portfolios, what are the signals indicating the presence of a temporary price deviation? Finally, given these signals, how should these assets be traded to optimize profitability given market conditions? Firstly, obtaining "similar" assets is ambiguous, and finding portfolios with long and short positions is not easy. Secondly, what information should

* Corresponding author.

E-mail addresses: gabriel.vergara@sansano.usm.cl (G. Vergara), werner.kristjanpoller@usm.cl (W. Kristjanpoller).

<https://doi.org/10.1016/j.asoc.2024.111255>

Received 11 July 2023; Received in revised form 4 December 2023; Accepted 3 January 2024

Available online 18 January 2024

1568-4946/© 2024 Elsevier B.V. All rights reserved.

be considered to determine patterns and the complex relationships that allow the identification of these market inefficiencies? Lastly, the optimal trading rule that maximizes the investor's objective given these complex market relationships. These challenges require flexible tools that utilize all available information and, naturally, deep learning techniques, which have proven to be extremely efficient in solving problems in high-dimensional spaces and finding complex relationships between variables. However, the problem at hand does not correspond to a typical problem of point prediction and categorization, but rather a comprehensive framework considering a horizon and a dynamic decision rule.

Considering these background aspects, this process is well represented as an online decision-making process as market conditions change. According to the requirements of the arbitrage scheme, deep reinforcement learning (DRL) functions as a direct adaptive optimal control system. The agent in DRL is the component that decides which actions to take to maximize the established reward. Therefore, the fundamental principle of DRL is to maximize the cumulative reward of the agent in the learning process, which leads to optimal decision-making strategies for different types of problems. Reinforcement learning has proven successful in various tasks, ranging from unmanned driving [2] and robot navigation [3] to playing video games. It has achieved the proposed tasks and, in some cases, surpassed human experts, such as AlphaZero defeating world champions in Shogi, Chess, and Go [4]. Hence, the question arises in the field of quantitative trading: Is it possible to train a DRL agent that can generate arbitrage strategies, obtain profits and reduce risk in a highly volatile and risky market like cryptocurrency?

This work proposes a unified framework that combines elements from classical statistical arbitrage theory and innovates in applying DRL to generate investment strategies. Fig. 1 represents the proposed model. The first module corresponds to the market training window; the processing module takes the market prices, makes the portfolios to trade, and the market signal representation (S_t). Then, the DRL module represents the DRL model used, and the agents make decisions through a policy network or a Q-value network as a Multilayer Perceptron. The Standard model is the benchmark model that works as a policy that predicts price direction directly.

Firstly, there is a module for generating arbitrage portfolios through cointegration. Then, signal extraction is performed based on the generated prices and portfolios. Finally, the DRL module takes these signals, processes them, and, based on the experience gained during training, makes optimal decisions to maximize the accumulated return. A fundamental difference is that the actions are not generated to maximize each individual action but within a given time horizon. Since the cointegration relationships among assets do not persist for long periods, they must be dynamically generated and used only for a short period of time.

A statistical perspective is used to construct the portfolios based on the cointegration relationship of price series. In this way, two portfolios of different assets replicating each other are built, creating market neutrality by taking opposing positions in each portfolio. Therefore, the difference in the value of these synthetic assets captures the temporary deviations in the prices of the underlying assets. Subsequently, various transformations and classic technical indicators are employed to detect patterns and obtain relevant information about the market state, considering aspects such as asset valuation, trend, volatility, and momentum. Thus, the market state is decomposed into different signals, which serve as input to the DRL agent or, in other words, represent the market state observed by the agent. Finally, to generate the optimal investment rule, the agent processes this market information using a neural network and makes decisions based on it to maximize the overall return of a trading session. Since each session involves a different arbitrage portfolio, generating a simulator of random scenarios on which the agent can be trained is necessary. For this purpose, a wide historical window is used, encompassing periods of extreme trend and volatility

and less volatile periods with no clear trend. The proposed scenarios correspond to past events, not synthetic data, as generating synthetic market data is not feasible. In this sense, training the agent involves maximizing the accumulated reward over a finite horizon, much like maximizing the score in a video game where random scenarios are generated.

The training was conducted using a training window from 2020-11-01 to 2022-10-11, comprising 34,000 records with a frequency of 30 min for 14 assets. Subsequently, the out-of-sample analysis was performed from 2022-10-19 to 2023-03-18, totaling 150 days of transactions. The results show, firstly, that DRL agents can generate strategies with positive returns in the out-of-sample period. These findings are particularly interesting as they demonstrate the success of an arbitrage strategy in an extremely risky and trending market, such as cryptocurrencies.

Secondly, the employed strategies significantly reduce risk, generating low-risk returns compared to the market. Furthermore, when considering different risk-adjusted performance metrics at daily and intraday trading levels, they achieve higher returns than the best-performing assets while maintaining lower risk levels.

Thirdly, when introducing transaction costs, the strategies maintain positive returns except for the Advantage Actor-Critic (A2C) agent, which performs worst. However, the Deep Q-Network (DQN) agent not only outperforms the standard COIN arbitrage strategy without transaction costs but also improves upon it when considering these costs while maintaining lower risk levels than the market.

Lastly, it is determined that the agent's actions are not random. The trained policies generate well-founded actions rather than errors, consistent with the results. Thus, it is established that generating an arbitrage strategy using DRL is possible, improving upon classical statistical methods.

The following structure of this paper is as follows: Section 2 presents a comprehensive investigation of related works. Section 3 introduces the formulated arbitrage model. First, defining the arbitrage portfolio construction. Second, describe the market signal representation. Third, the DRL agent or arbitrageur is described, and finally, the training method is discussed. Section 4 analyzes the results obtained in different aspects, conducting a daily and intraday analysis of strategy risk along with risk-adjusted performance, and finally studying the agent's behavior. Lastly, Section 5 concludes the work.

2. Related work

Recently, machine learning, particularly deep learning, has become an increasingly popular tool and technique for various applications in finance, including portfolio selection problems. The increasing availability of data and growing computer processing power have contributed to the popularity of deep learning. Reinforcement learning (RL) is a branch of machine learning based on behaviorist psychology. RL aims to learn how to achieve a goal, such as automated control or sequential decision-making, by interacting with the environment through an adaptive learning process.

Neuneier (1997) [5] formulated financial markets as Markov Decision Process (MDP) under several assumptions and simplifications regarding market characteristics. Such formulation is crucial because finite MDPs enable Q-learning and RL techniques to converge on optimal policy. Furthermore, developing DRL techniques enables the automatic learning of more sophisticated behaviors. DRL can achieve outstanding performance of various realistic tasks through automatic engineering and end-to-end learning through gradient descent [6].

For the latest financial applications, Deng et al. (2016) [7] introduced a trading system incorporating a deep direct RL framework with a fuzzy representation of past returns, which greatly reduced market uncertainties, increasing the feasibility of extracting more complex features. A portfolio management framework was designed by using the EIIE topology [8], which was used to reveal the specific characteristics

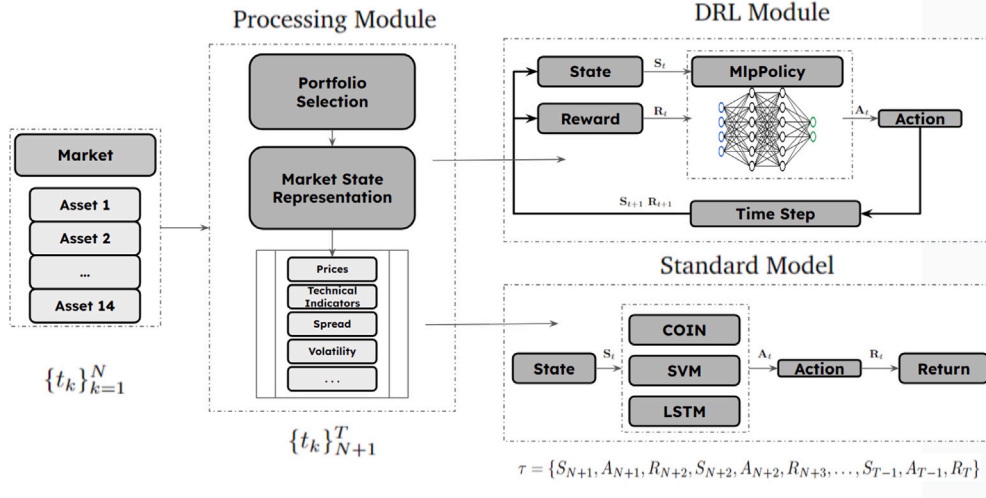


Fig. 1. Proposed Framework model.

of individual assets without making biased decisions affected by any long-past poor records of particular assets. Neves et al. (2018) [9] generated a short-term speculation system in the foreign exchange market based on reinforcement learning. Liang et al. (2018) [10] implement three state-of-the-art RL algorithms for portfolio management: Deep Deterministic Policy Gradient (DDPG), Proximal Policy Optimization (PPO), and Policy Gradient (PG). They also propose an adversarial training method that can greatly improve training efficiency. Lee et al. (2020) [11] propose the MultiAgent RL-based portfolio management System (MAPS) and devise a new loss function with a diversification penalty term to effectively encourage agents to maximize both diversity and returns. Wu et al. (2020) [12] introduced an adaptive trading method based on DRL using the Gated Deep Q-learning (GDQN) and Gated Deterministic Policy Gradient (GDPG) approaches, showing positive results. Zhang et al. (2020) [13] compared DRL in various markets using a reward function that incorporates a target conditional volatility level, achieving results that outperform classical methods and maintaining profits under high transaction costs. Liu et al. (2021) [14] proposed an automatic high-frequency transaction framework based on DRL using a Long Short-term Memory Network (LSTM) as the policy based on a PPO algorithm on bitcoin. They compare standard prediction models, including Support Vector Machine (SVM), Multi-Layer Perceptron (MLP), LSTM, Temporal Convolutional Network (TCN), and Transformer, and conclude that LSTM is the best fit to use as the policy function. The empirical studies confirm that the proposed method performs Superior to various common trading strategy benchmarks for a single financial product. Chen et al. (2021) [15] proposed a multimodal reinforcement trading system based on sentiment analysis, which includes a deep neural network and multimodal deep recurrent neural network where the multimodal integration of price information with news sentiment enables the agent to make profits. Lin et al. (2022) [16] proposed a novel Multiagent-based DRL for risk-shifting portfolio management (MABDRL) with outstanding performance on several common metrics with a 37% Sharpe ratio, where each agent is equipped with sophisticated deep policy networks that enable the proposed RL agent to learn risk-shifting behaviors with stable convergence.

On the other hand, applications of DRL in arbitrage or pairs trading are rather scarce. Mulvey et al. (2020) [17] and Kim and Kim (2019) [18] employed machine learning methods in parametric statistical arbitrage models. In contrast, Pelger et al. (2021) [19] generalized the arbitrage model by incorporating principal component analysis, instrumented principal component analysis (PCA and IPCA) and alpha factors models. They combined these arbitrage portfolios with convolutional networks, transformers, Fourier analysis, feed-forward networks, and the parametric Ornstein-Uhlenbeck model with thresholds for optimal investment decision generation.

3. Model formulation

This section details the model formulation and the proposed framework step-wise.

1. Portfolio Definition:

Portfolio construction is made through cointegration, using a past time horizon of 288-time steps (6 days) to estimate the cointegration portfolios. The resulting portfolios will be traded using a time horizon of 48 steps (1 day).

2. Market Representation:

After the portfolios are defined, the signals that define the market are made using the market prices and the portfolios through different signals representing market characteristics.

3. Taking Positions:

With all the previous steps done, the last component defines how the trading decision is made. The Markov Decision Process is explained. Then, the DRL algorithms are briefly detailed, and finally, how is the Agent trained?

This section's main goal is to present the agent's action and how the agent is trained to choose this investment decision through a policy. In the policy network, the policy is represented as a Multi-Layer Perceptron that outputs the action directly, while the Q-Network uses the Mlp as the Q-value representation that outputs the action that maximizes the Q-value.

First, the portfolio construction is made through cointegration. Second, the market representation is formulated through signals from different technical indicators, volatility, and portfolio spreads. Third, the decision is made through a MlpPolicy that outputs the action.

3.1. Arbitrage portfolios

The portfolio allocation is essential to define an investment strategy. In this way, constructing an arbitrage portfolio is a decision to improve the risk-return performance of the investment. The idea is to model a long-term asset return behavior and invest in long and short positions in a neutral portfolio. The arbitrage portfolios must be built by applying econometric models and the following steps. Considering a set of normalized prices $P_{n,t}$ of assets $n = 1, \dots, N$ at time $t = 1, \dots, T$. To generate the arbitrage portfolios, we assume an Error Correction Model (ECM) for the log prices where $\ln(P_{n,t}) = p_{n,t}$ denoted as $\mathbf{p}_t = (p_{1,t}, \dots, p_{n,t})'$. Considering a $VAR(k)$ model on $\mathbf{p}_t$ which can be rewritten as:

$$\Delta \mathbf{p}_t = \mu + \sum_{i=1}^{k-1} \Gamma_i \Delta \mathbf{p}_{t-i} + \Pi \mathbf{p}_{t-1} + \varepsilon_t \quad (1)$$

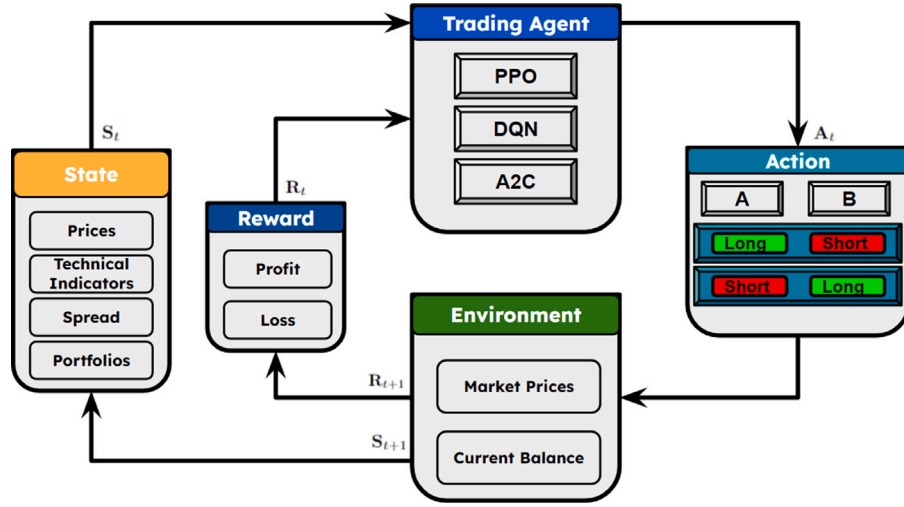


Fig. 2. Interaction scheme of the deep reinforcement learning system for the trading application.

where $\Gamma_i \in \mathbb{R}^{n \times n}$, $\Pi \in \mathbb{R}^{n \times n}$ and $\varepsilon_t \stackrel{iid}{\sim} N(0, A)$. If $\text{rank}(\Pi) = n$, then $\mathbf{p}_t$ is stationary. If $\text{rank}(\Pi) = 0$ then $\Pi = 0$ implying that $\Delta \mathbf{p}_t$ is a $VAR(k-1)$ process and there are no cointegration vectors found. If $1 \leq \text{rank}(\Pi) = r \leq n-1$, then there exist $n \times r$ matrices with rank r , $\mathbf{A}$ and $\mathbf{B}$, such that Π in model (1) can be expressed as:

$$\Pi = \mathbf{A}\mathbf{B}' \quad (2)$$

and $\mathbf{b}_1'\mathbf{p}_t, \dots, \mathbf{b}_r'\mathbf{p}_t$ are stationary, where $\mathbf{B} = (\mathbf{b}_1, \dots, \mathbf{b}_r)$. The Model (1) with Π satisfying $\Pi = \mathbf{A}\mathbf{B}'$ is the ECM, with cointegration vectors $\mathbf{b}_1, \dots, \mathbf{b}_r$. Then, from the $\mathbf{p}_t$ as the cumulative total log return of each asset, the r columns of $\mathbf{B}$ can be used to form r cointegrated (COIN) portfolios. However, these portfolios are not necessarily market-neutral.

To generate the arbitrage portfolio, each cointegration vector is normalized to make a dollar-neutral portfolio. Given the cointegration vector $\mathbf{b}_i = (b_i^1, \dots, b_i^n)'$ defining the long and short positions for every asset:

$$k \in L_i \iff b_i^k \geq 0, \quad \forall i = 1, \dots, r \quad (3)$$

$$k \in S_i \iff b_i^k < 0, \quad \forall i = 1, \dots, r \quad (4)$$

Then, the sets L_i and S_i define the directional position amount of each asset of the cointegrated portfolio formed by the cointegration vector $\mathbf{b}_i$.

The long portfolio is given by the assets where the cointegration coefficient is positive and the short portfolios by the negative coefficients. Therefore, r dollar neutral portfolios are built. For each vector $\mathbf{b}_i$, the corresponding normalization constant are $l_i = \sum_{k \in L_i} |b_i^k|$ and $s_i = \sum_{k \in S_i} |b_i^k|$. So, the weight of each asset in the portfolio is given by :

$$W_i^{(k)} = \begin{cases} b_i^{(k)}/l_i & \text{if } b_i^{(k)} \geq 0 \\ b_i^{(k)}/s_i & \text{if } b_i^{(k)} < 0 \end{cases} \quad (5)$$

These r portfolios are dollar-neutral, and the arbitrage portfolio $\mathbf{P}$ is built by a simple average of all the portfolios:

$$\mathbf{P} = \frac{1}{r} \sum_{k=1}^r \mathbf{W}_k, \quad \mathbf{W}_k = (W_k^1, \dots, W_k^n) \quad (6)$$

Finally, the portfolio formed by the positive weights will be denoted as portfolio $\mathbf{A}$, and the portfolio formed by negative weights will be denoted portfolio $\mathbf{B}$ so that $\mathbf{P} = \mathbf{A} - \mathbf{B}$.

3.2. Signal generation

With the corresponding arbitrage portfolio defined, the next step is to define the signals extracted based on market information. These

signals are the input for the trading policy; in other words, they define the market state observed by the agent (the algorithm).

1. Candlesticks

Considering the weights of the portfolios, we have 30-minute candlestick data for each asset in the interval. Thus, the *Open*, *High*, *Low*, *Close* (OHLC), and *Volume* values are constructed for assets A and B. The values are normalized based on the closing price of the first candle, which is set to 1, so the *Open*, *High*, and *Low* values of the first candle do not necessarily start at 1.

Considering these values, initially, eight states are defined, considering the normalized values for OHLC, $\mathbf{O}_t = (O_t^1, \dots, O_t^N)'$, and similarly for the others:

$$\begin{aligned} O_t^i &= \text{Open}_t^i / \text{Close}_0^i, & H_t^i &= \text{High}_t^i / \text{Close}_0^i, \\ L_t^i &= \text{Low}_t^i / \text{Close}_0^i, & C_t^i &= \text{Close}_t^i / \text{Close}_0^i, \forall i \end{aligned} \quad (7)$$

Considering the weights $W_A = (w_A^1, \dots, w_A^N)'$ and $W_B = (w_B^1, \dots, w_B^N)'$, where $\mathbf{P} = W_A - W_B$, we obtain the OHLC values for portfolios A and B:

$$O_t^A = \mathbf{O}_t W_A', \quad H_t^A = \mathbf{H}_t W_A', \quad L_t^A = \mathbf{L}_t W_A', \quad C_t^A = \mathbf{C}_t W_A' \quad (8)$$

$$O_t^B = \mathbf{O}_t W_B', \quad H_t^B = \mathbf{H}_t W_B', \quad L_t^B = \mathbf{L}_t W_B', \quad C_t^B = \mathbf{C}_t W_B' \quad (9)$$

These series are used to construct other indicators; however, only the closing series are considered a state signal.

2. Moving Average Convergence Divergence

The Moving Average Convergence Divergence (MACD) is a popular technical indicator commonly used in stock trading to identify momentum, trend, and price strength changes. The MACD is calculated by subtracting the 26-period exponential moving average (EMA) from the 12-period EMA and then plotting a 9-period EMA of the MACD as a signal line.

$$\text{MACD}(l, s)_t = \text{EMA}(l)_t - \text{EMA}(s)_t \quad (10)$$

where

$$\text{EMA}(k)_t = \text{Price}_t k + \text{EMA}_{t-1}(1-k) \quad (11)$$

This indicator is used with a long-term average of 14 periods and a short-term average of 5 periods. It will be calculated for the closing prices of portfolios A and B.

3. Relative Strength Index

The relative strength index, or RSI, is computed through the average of the upward and downward price movements:

$$RSI(N) = 100 - 100 / \left(1 + \frac{\sum_{k=t-N}^N [S_k - S_{k-1}]^+}{\sum_{k=t-N}^N [S_k - S_{k-1}]^-} \right) \quad (12)$$

This indicator is used with $N = 20$ based on the closing price S_t . It is calculated for portfolios A and B, resulting in 2 additional series.

4. Dynamic Moving Average

A Kalman filter generates a moving average signal of the price for synthetic assets A and B. This filter models the price series S_t by considering a not observable moving average μ_t in the following way:

$$\text{Measurement Equation: } S_t = \mu_t + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma^2) \quad (13)$$

$$\text{State Equation: } \mu_t = \mu_{t-1} + v_t, \quad v_t \sim N(0, \sigma_\mu^2) \quad (14)$$

Additionally, the Kalman gain is obtained:

$$K_t = \frac{\text{Cov}[\mu_t, v_t | \mathcal{F}_{t-1}]}{\mathbb{V}[v_t | \mathcal{F}_{t-1}]} = \frac{\text{Cov}[\mu_t, v_t | \mathcal{F}_{t-1}]}{\mathbb{V}[\mu_t | \mathcal{F}_{t-1}] + \mathbb{V}[\varepsilon_t | \mathcal{F}_{t-1}]}, \quad v_t = S_t - \mathbb{E}[S_t | \mathcal{F}_{t-1}] \quad (15)$$

Then, the conditional expectation of μ_t given the price S_t is:

$$\hat{\mu}_t = \mathbb{E}[\mu_t | S_t] = (1 - K_t) \mathbb{E}[\mu_t | \mathcal{F}_{t-1}] + K_t S_t \quad (16)$$

Where the volatilities are estimated in the training sample, more details about filters can be found in Durbin & Koopman 2012 [20].

5. Stock Strength

The *Stock Strength* indicator proposed in Wu, Chen, Wang, Troiano, Loia, and Fujita 2020 [12] aims to discriminate whether the market is in a trending or volatile regime:

$$ST = \frac{S_n - S_1}{\sum_{i=1}^n |S_{i+1} - S_i|} \quad (17)$$

It is applied to a window of 10 intervals for synthetic assets A and B. This indicator measures the accumulated return over an interval of n periods divided by the sum of the absolute returns for the period. Therefore, if the market is volatile, this indicator will be close to zero, while in a trending regime, it will be either strongly positive or strongly negative.

6. Exponentially Weighted Moving Average

The conditional volatility is calculated using the standard EWMA for the return series of synthetic assets A and B, along with the spread series A-B, to introduce dependence on conditional volatility.

$$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_t^2 \quad (18)$$

A memory parameter $\lambda = 0.94$ is used, which means that a volatility shock has a high persistence over time.

7. Bias

The Bias measures the relative deviation of the price from a reference moving average value. In this case, it is applied to the moving average obtained through the Kalman filter.

$$\text{Bias}_t = \frac{S_t - \mu_t}{\mu_t} \quad (19)$$

This measure is applied to synthetic assets A and B.

8. Signs

Another measure of relative strength is the sign of the returns. Unlike the previous measures, it does not measure the net effect or quantitative strength comparison based on returns. Instead, it counts the number of positive returns versus negative returns.

$$\text{Signs}_t = \frac{\sum_{i=0}^N [\text{sign}(r_{t-i})]^+}{N} \quad (20)$$

This measure is applied to synthetic assets A and B, using a window of 10 periods. Unlike other measures, it takes a discrete set of values.

9. Stochastic Relative Strength Index

Additionally, the stochastic RSI, a scaled version of the regular RSI using a scaler based on the minimum and maximum values over a certain period, is applied.

$$\text{SRSI}_t = \frac{\text{RSI}_t - \min_{k=t-N:t} \text{RSI}_k}{\max_{k=t-N:t} \text{RSI}_k - \min_{k=t-N:t} \text{RSI}_k} \quad (21)$$

This is applied to synthetic assets A and B using a window of 20 periods.

10. Candle Shadows

Finally, a metric of price volatility concerning the range of the candlesticks in the period is applied.

$$\text{Upper}_t = \frac{\text{High}_t - \max\{\text{Open}_t, \text{Close}_t\}}{\text{High}_t - \text{Low}_t} \quad (22)$$

$$\text{Lower}_t = \frac{\min\{\text{Open}_t, \text{Close}_t\} - \text{Low}_t}{\text{High}_t - \text{Low}_t} \quad (23)$$

These measures are applied to the synthetic assets A and B.

Additionally, the spread A - B is introduced as a state. The moving average calculated through the Kalman filter for the spread is also used as a state and, finally, a market turbulence index, which corresponds to the Mahalanobis distance of market returns:

$$\text{Turbulence} : T_t = \sqrt{(\mathbf{r}_t - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{r}_t - \boldsymbol{\mu})} \quad (24)$$

where $\boldsymbol{\mu}$ corresponds to the vector of mean returns during the training period, $\mathbf{r}_t$ represents the returns of the assets, and $\boldsymbol{\Sigma}$ is the covariance matrix of the returns during the training period. This brings the total number of states to 26, which is expected to provide the agent with information about market interactions for learning purposes.

3.3. Trading decision

Fig. 2 represents the interaction scheme of the deep reinforcement learning system for the trading application. The states correspond to those defined through technical indicators, volatilities, and others generated from synthetic assets and spreads. The possible actions are either taking a long position in A and a short position in B or taking a short position in A and a long position in B.

3.3.1. Markov decision process

The trading problem can be formulated as a Markov Decision Process (MDP) where an agent interacts with the environment at discrete time steps. At every moment t , the agent observes the environment state S_t . Given the observed state, S_t , the agent chooses an action A_t , and based on the action, it gets a reward R_{t+1} on the next time step, and the agent observes a new state S_{t+1} . The interaction between the agent and the environment produces a trajectory of state, actions, and rewards $\tau = (S_0, A_0, R_1, S_1, A_1, R_2, \dots)$. The objective of the agent is to maximize the expected return at any given time denoted as G_t :

$$G_t = \sum_{k=t+1}^T \gamma^{k-t-1} R_k \quad (25)$$

where γ is the discount factor.

The state is represented as the signals extracted from the arbitrage portfolios and spreads. The reward function is the discounted total cumulative reward with a discount factor of 0.995. The action space is discrete, as the portfolios are premed; the agent does not influence the portfolio composition; it only has to decide which side to take. Given the two arbitrage portfolios A and B, the actions are long-A and short-B ($A_t = 1$) or short-A and long-B ($A_t = -1$), so there are only two possible actions to take, $A_t \in \{-1, 1\}$. Given the portfolio returns $r_t^A = \ln P_t^A - \ln P_{t-1}^A$ and $r_t^B = \ln P_t^B - \ln P_{t-1}^B$, the reward or return of the action A_{t-1} is given by

$$R_t = A_{t-1} \left[e^{r_t^A} - e^{r_t^B} \right]. \quad (26)$$

Finally, the last fundamental aspect of the model corresponds to the policy π . The policy governs the actions to be taken, given the state of the environment. It is a function that maps the state space to a probability distribution over the action space (or deterministic). Furthermore, the policy is stationary, meaning it is time-invariant, i.e., $\pi(a|s_t) = \pi(a|s_{t+h})$ if $s_t = s_{t+h}$ for all h . A deterministic policy is denoted as $\pi(s)$, where only a single action is selected for each state.

As the return of a trajectory corresponds to a realization and both the environment and the policy are stochastic, for a given policy and initial state, the trajectory represents a realization of a stochastic process. The aim is to quantify the expected return, which is called value. To do so, we define the state value and state-action value.

The state value $V(s)$ is defined as the expected return when the agent starts in state s and acts according to policy π :

$$V^\pi(s) = \mathbb{E} \left[\sum_{h=0}^{\infty} \gamma^h R_{t+h} | s_t = s \right] \quad (27)$$

For the state-action pair (s, a) , we define the value $Q^\pi(s, a)$, which, unlike the state value function $V^\pi(s)$, depends on the action a :

$$Q^\pi(s, a) = \mathbb{E} \left[\sum_{h=0}^{\infty} \gamma^h R_{t+h} | s_t = s, a_t = a \right] \quad (28)$$

3.3.2. DRL algorithms

Deep Q Network (DQN)

Deep Q Networks (Mnih et al. 2013 [21], Mnih et al. 2015 [22]) approximates the Q value function to estimate the expected reward of the agent for a given state-action. This function approximation is made as a neural network with parameters θ . So with the Q-Value approximation Q_θ , we minimize the mean square error between the current and target Q to derive the optimal state-action value function:

$$L(\theta) = \mathbb{E} [(Q_\theta(S, A) - Q'_\theta(S, A))^2] \quad (29)$$

$$Q'_\theta(S_t, A_t) = r + \gamma \arg \max_{A'} Q_\theta(S_{t+1}, A_{t+1}) \quad (30)$$

with the objective function $L(\theta)$. This is the general framework of DQN algorithms,

Policy optimization

Proximal Policy Optimization (PPO) algorithm (Schulman et al. 2017 [23]) is a policy gradient algorithm that aims to maximize the expected cumulative reward by optimizing the policy directly. Assuming the θ the parameters of the network or the policy $\pi_\theta(A, S)$. Maximizing the expected cumulative reward $J(\theta)$ with a gradient ascent to learn θ :

$$J(\theta) = \mathbb{E} \left[\sum_{t=0}^{T-1} R_{t+1} | \pi_\theta \right] \quad (31)$$

$$\nabla_\theta J(\theta) = \sum_{t=0}^{T-1} \nabla_\theta \ln[\pi_\theta(A_t | S_t)] G_t \quad (32)$$

where G_t is the expected discounted cumulative rewards (25). Policy gradient methods directly learn the policy and can output a probability distribution over the actions. These methods are useful for designing stochastic policies or continuous action spaces.

Advantage actor-critic (A2C)

Advantage Actor-Critic (Mnih et al. 2016 [24]) The A2C method is proposed to solve the training problem from policy gradient methods by updating the policy in real-time. There are two networks involved: one is the actor-network that outputs the policy, and the other network is the critic component that measures how good the chosen action is given the state. We can update the policy network $\pi(A|S, \theta)$ by maximizing the objective function:

$$J(\theta) = \mathbb{E} [\ln[\pi(A|S, \theta)] A_{adv}(S, A)] \quad (33)$$

where $A_{adv}(S, A)$ is the advantage function defined as:

$$A_{adv}(S_t, A_t) = R_t + \gamma V(S_{t+1}|w) - V(S_t|w) \quad (34)$$

To compute the advantages of another network, the critic network with parameters w to model the state value $V(s|w)$ is used. Therefore, the update of the critic network is done by gradient descent to minimize the temporal difference error:

$$J(w) = (R_t + \gamma V(S_{t+1}|w) - V(S_t|w))^2 \quad (35)$$

The A2C is useful in continuous action spaces as the use of the advantage function constrains policy variance and update the policy in real-time.

3.4. Training configuration

The training process is as follows: First, define the arbitrage scenario

- Step 1: define a time window of length W of historical data to estimate the cointegration portfolios.
- Step 2: use the estimated cointegration vector and the historical data to construct the arbitrage portfolios **A** and **B**. The Johansen Cointegration rank test chooses the cointegration rank at a confidence of 90%.
- Step 3: with the historical data and arbitrage portfolios, compute the environment states.
- Step 4: define a trading horizon of length T to trade the arbitrage portfolios.

A historical candle window with a frequency of 30 min was used for scenario generation. The training period starts from 2020-11-01 00:00:00 until 2022-10-11 05:30:00, comprising 34,000 price candles for the 14 assets under study. In this way, a window of 336 time intervals is used to generate a trading scenario. This interval is divided into two components: the first 288 values are used for portfolio construction and states, and subsequently, starting from the last value, there are 48 intervals to make 47 decisions, resulting in over 30,000 distinct scenarios. Fig. 3 shows the training and testing scenario segmentation scheme. In the training set, scenarios overlap, while in the testing set, overlap does not occur, and no information used for training is present.

Subsequently, the testing period starts on 2022-10-19 at 07:30:00 and ends on 2023-03-18 at 07:30:00, completing a total of 150 days of transactions.

To train the DRL models, Python 3.8 programming language is used. The agents, namely Advantage Actor-Critic (A2C) (Mnih et al. 2016 [24]), Deep Q Network (DQN) (Mnih et al. 2013 [21]; Mnih et al. 2015 [22]), and Proximal Policy Optimization (PPO) (Schulman et al. 2017 [23]), are trained on the Stable Baselines3 library.

A gamma factor of 0.995 was used, and the MlpPolicy was employed, corresponding to a fully connected dense feed-forward neural network with ReLu activation functions that return action probabilities.

Regarding training quantity, a total of 1e6 steps were trained, with each scenario consisting of 47 steps. This means that approximately 21,000 scenarios were trained on. Additionally, the scenarios were randomly selected from the sample with equal probability.

4. Empirical analysis

The results are presented considering the daily performance of the strategies as a result of the intra-daily decision-making process. In addition, risk and risk-adjusted return measurements are performed using different metrics.

Value-at-Risk (VaR) and Expected Shortfall (ES) risk measures consider tail risks or extreme losses incurred at a 5%

Standard deviation is a popular and easily understandable risk measure. It represents the level of variability or scale at which returns are obtained. While it accurately measures variability, it is easily

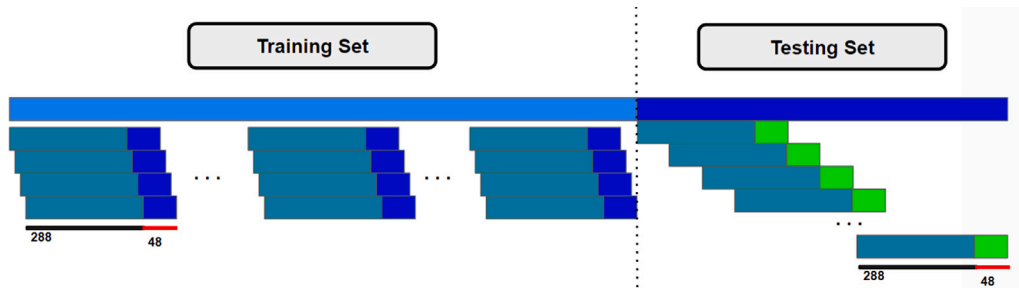


Fig. 3. Training and testing scenario segmentation scheme.

Table 1

Summary statistics of the sample of logarithmic returns.

	SOL	AVAX	ALGO	CAKE	AAVE	DOT	UNI	ETH	ADA	BNB	XRP	LTC	BCH	BTC
mean	-0.5	-0.4	-0.5	-0.3	-0.3	-0.4	-0.2	-0.2	-0.3	-0.1	-0.2	-0.1	-0.3	-0.1
std	6.5	5.8	5.1	4.6	6.1	4.8	5.3	4.3	4.7	3.7	4.2	4.5	4.3	3.3
min	-54.9	-35.8	-26.2	-31.5	-24.3	-22.3	-21.4	-19.1	-20.7	-20.3	-20.7	-20.9	-18.3	-16.7
50%	-0.5	-0.3	0.3	0.0	-0.1	-0.2	0.0	-0.1	-0.2	-0.1	0.0	-0.1	0.0	-0.1
max	28.2	22.0	18.9	18.9	28.4	18.3	18.7	16.6	16.7	12.9	21.1	17.7	20.1	13.5
VaR 5%	-10.1	-10.0	-8.6	-6.9	-10.6	-8.0	-9.1	-7.2	-7.9	-6.0	-6.5	-7.6	-7.5	-5.4
ES 5%	-15.7	-14.3	-13.1	-12.8	-14.5	-12.0	-12.5	-11.1	-11.7	-9.4	-10.0	-11.0	-11.3	-8.7
skew	-1.23	-0.61	-0.92	-1.25	-0.07	-0.49	-0.20	-0.38	-0.23	-0.85	0.05	-0.36	-0.35	-0.47
kurt	12.36	4.12	3.85	7.37	2.52	2.88	1.49	2.91	2.37	4.65	5.17	2.51	3.18	4.30

influenced by positive returns, particularly extreme positive returns, which do not necessarily indicate higher risk.

Another relevant measurement is Drawdown, which measures peak-to-peak losses, i.e., accumulated losses from the portfolio's highest value. This measurement is crucial when evaluating strategies in extreme risk environments, as it measures the percentage amplitude of the generated losses.

Finally, considering that strategies may differ in their higher moments, which can determine whether one strategy is better than another in generating profitability relative to the associated risk, they are measured through the Aumann-Serrano coefficient, AS. The risk-adjusted performance is then compared by considering the four moments of the distribution through the economic performance measure, EPM.

4.1. Data

Regarding the assets used, studying their performance from 2022-01-01 to the date 2023-03-17, encompassing a total of 440 trading days with a daily granularity. A summary of statistics is presented in Table 1 for a total of 14 assets: Bitcoin (BTC), Ethereum (ETH), BNB, Ripple (XRP), Cardano (ADA), Solana (SOL), Polkadot (DOT), Bitcoin cash (BCH), Litecoin (LTC), Avalanche (AVAX), Algorand (ALGO), AAVE, UniSwap (UNI), PancakeSwap (CAKE).

Table 1 shows the descriptive statistics of the sample of 440 daily logarithmic returns. The Value at Risk and Expected Shortfall are derived from the empirical distribution of the obtained data. The asset with the highest daily loss during the period is Solana (SOL), with a minimum return of -0.549, corresponding to a one-day loss of 42.25% in percentage terms. On the other hand, Bitcoin (BTC) has the lowest maximum loss with a one-day maximum loss of 15.38%. As for the maximum gains, Aave (AAVE) has the highest one-day maximum gain at 32.79%, while BNB has the lowest maximum gain at 13.73%.

It is observed that during the period, all assets have a negative average return, and they are all negatively skewed except for Ripple (XRP), which has a positive skewness coefficient of 0.05. Regarding the excess kurtosis values, all assets exhibit some level of excess kurtosis, but Solana stands out with an excess of 14.

The asset with the lowest correlation is Ripple (XRP) with PancakeSwap (CAKE), with a correlation of 69%. The assets show a high

correlation, as observed in the previous graphs. This makes achieving diversification and reducing systemic risk challenging when constructing portfolios with long positions in these assets. Additionally, considering the high volatility in these assets, it creates a suitable market for arbitrage activities in short intervals, maximizing profitability based on short-term inefficiencies.

Fig. 4 shows the performance of the 14 assets considering their values normalized to the closing price on 2022-01-01 at 00:00 h. This corresponds to a sample of a total of 440 trading days.

In both charts, both in the testing period (Fig. 5) and in the broader market view (Fig. 4), it can be observed that although the market is highly correlated, in the very short term, the normalized price series of all assets exhibit a coordinated movement that later begins to diverge. However, they still maintain the same overall direction of movement, considering that at specific moments, an asset deviates from the market trend and remains higher or lower. This reaffirms the potential of statistical arbitrage strategies to generate returns outside the market trend in a highly risky environment.

4.2. Risk and performance management

The Sharpe ratio is one of the most popular performance metrics in which the expected return is adjusted against another alternative, usually a risk-free rate or, more generally, the expected return of a low-risk alternative investment, over the standard deviation of the return. This metric assumes that the distribution of returns can be categorized into the family of location and scale distributions. Therefore, the fundamental elements of this metric correspond to a location parameter (the mean) and a scale parameter (standard deviation) that serves as a measure of risk. However, a distribution has more characteristics than just its location and scale. These include higher-order moments such as skewness and kurtosis (shape parameter), which are equally important. It is known that returns exhibit characteristics in these higher moments; they are negatively skewed and have high kurtosis, greater than a normal distribution. Therefore, since the Sharpe ratio only considers the first two moments, it ignores the characteristics of the higher moments. For this reason, a metric that considers these elements is proposed.

In contrast to the Sharpe ratio, the proposed metric seeks strict monotonicity with respect to stochastic dominance, so it cannot be

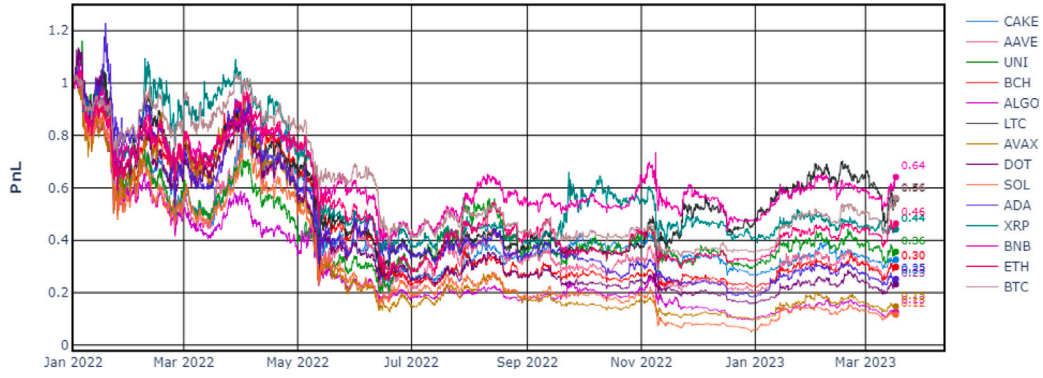


Fig. 4. Performance of the analyzed assets.

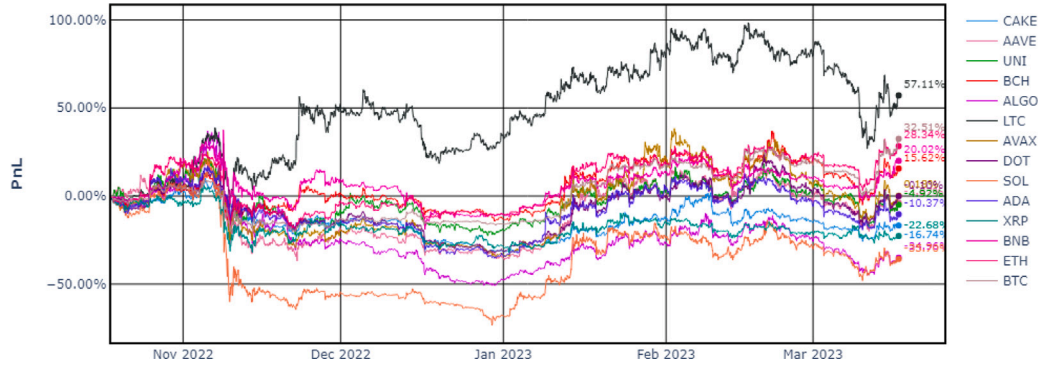


Fig. 5. Cumulative returns for the selected assets in the 150 days of testing period.

established solely based on the location and scale of the distribution. The proposed Economic Performance Measure (EPM) metric is obtained by dividing the expected value of an investment opportunity by its Aumann & Serrano (2008) economic risk index [25], Ulrich & Pigorsch (2012) [26], denoted as the AS index. If the returns are normally distributed, then the metric coincides with the Sharpe ratio in terms of order (if $r \sim F \rightarrow N(\mu, \sigma^2)$, then $EPM \rightarrow 2\text{Sharpe}^2$). Additionally, this metric is suitable for both high and low frequencies, while the Sharpe ratio is more appropriate for low frequencies. To this end, parametric and non-parametric moment estimators are proposed for estimating the EPM. For parametric estimation, it is proposed that the returns follow a generalized hyperbolic distribution (GHYP), which can model asymmetry and heavy tails. Thus, higher-order moments are considered for the EPM metric in a straightforward and interpretable manner. One of the metric results is that for returns with relatively high skewness and low kurtosis, the metric establishes a higher order than the Sharpe ratio.

To contrast the different implemented strategies, various performance criteria will be used. Firstly, the final profitability of the strategy (PnL) is recorded, followed by risk-reward metrics such as the Sharpe ratio, the Calmar ratio, and the Economic Performance Measure (EPM) using a generalized hyperbolic distribution.

4.2.1. Economic performance measure

Considering $\tilde{r} \sim F$ and r^f the risk-free rate, then $r = \tilde{r} - r^f \sim F$ corresponds to the excess return. The Economic Performance Measure (EPM) metric relative to the AS risk index is defined as follows:

$$EPM(r) = \frac{E(r)}{AS(r)} = \frac{E(\tilde{r}) - r^f}{AS(\tilde{r} - r^f)} \quad (36)$$

Considering $M(t)$ as the Moment Generating Function (MGF) of the returns, the AS index is the value $s > 0$ such that $M(-1/s) = 1$, and

we denote it as $AS(r) = \{s > 0 : M(-1/s) = 1\}$. Assuming that a Generalized Hyperbolic Distribution generates r :

$$r \sim GH(\lambda, \chi, \psi, \mu, \sigma^2, \gamma), \quad r \stackrel{d}{=} \mu + W\gamma + \sqrt{W}AZ \quad (37)$$

Where $W \sim GIG(\lambda, \chi, \psi)$ (Generalized Inverse Gaussian), $Z \sim N(0, 1)$. The parameters λ, χ, ψ determine the shape of the distribution, i.e., how much weight is assigned to the tails and the center. Generally, the distribution approaches a normal distribution for large values of these parameters. Meanwhile, μ corresponds to the location parameter, and σ^2 corresponds to the scale parameter. Finally, the parameter γ corresponds to the skewness parameter, where $\gamma = 0$ represents a symmetric distribution. Thus, the stochastic representation is given as follows:

$$r|W \sim N(\mu + W\gamma, W\sigma^2) \quad (38)$$

Then we have that $E(r) = \mu + E(W)\gamma$ and $Var(r) = \gamma^2 Var(W) + E(W)\sigma^2$. Next, the probability density function is given by:

$$f_r(x) = \int_0^\infty f_{r|W}(x|w)f_W(w)dw = \int_0^\infty \frac{e^{\frac{(x-\mu)\gamma}{\sigma^2}}}{\sqrt{2\pi\sigma^2 w}} \exp\left\{-\frac{Q(x)}{2w} - \frac{w\gamma^2}{2\sigma^2}\right\} f_W(w)dw \quad (39)$$

$$f_r(x) = \frac{(\sqrt{\psi/\chi})^\lambda (\psi + \gamma^2/\sigma^2)^{1/2-\lambda}}{\sqrt{2\pi\sigma^2} K_\lambda(\sqrt{\chi\psi})} \times \frac{K_{\lambda-1/2}\left(\sqrt{(\chi + Q(x))(\psi + \gamma^2/\sigma^2)}\right) e^{\frac{\gamma(x-\mu)}{\sigma^2}}}{\left(\sqrt{(\chi + Q(x))(\psi + \gamma^2/\sigma^2)}\right)^{1/2-\lambda}} \quad (40)$$

where $Q(x) = (x - \mu)^2/\sigma^2$ and $K_\lambda(\cdot)$ is a modified Bessel function of the third kind. The moment generating function (MGF) can be easily obtained since it is a mixture of normal distributions, thus:

$$\mathbf{M}_{GH}(t) = E[e^{t'W}] = e^{t'\mu} E(\exp(W(t\gamma + 1/2t^2\sigma^2))) \quad (41)$$

$$= e^{t'\mu} \left(\frac{\psi}{\psi - 2t\gamma - t^2\sigma^2} \right)^{\lambda/2} \frac{\mathbf{K}_\lambda(\sqrt{\psi(\chi - 2t\gamma t^2\sigma^2)})}{\mathbf{K}_\lambda(\sqrt{\chi\psi})}, \quad \chi \geq 2t\gamma + t^2\sigma^2 \quad (42)$$

Following the EPM using a NIG, the GH has the particular case that it coincides with the NIG when $\lambda = -1/2$. Then, following Tiantian Li, Young Shin Kim, Qi Fan, and Fumin Zhu (2021) [27], defining the Aumann-Serrano coefficient $R_{AS}(X)$ as the solution to $E(\exp(-X/R)) = 1$, we have that $R_{AS}(r) = \sigma^2/(\gamma + \sqrt{\gamma^2 + \sigma^2\psi})$. Therefore:

$$\text{EPM}_{GH}(r) = \frac{E(r)}{R_{AS}(r)} = \frac{E(r)(\gamma + \sqrt{\gamma^2 + \sigma^2\psi})}{\sigma^2} \quad (43)$$

4.2.2. Maximal drawdown

Considering $W(t)$ as the total wealth at time t , the drawdown is the accumulated loss from the peak wealth until the current time t , so it measures the current loss concerning the maximum wealth achieved.

Therefore, the maximum drawdown is defined as:

$$\text{MDD} = \max_{0 \leq \tau \leq T} \left\{ \max_{0 \leq t \leq \tau} \frac{W(t) - W(\tau)}{W(t)} \right\} \quad (44)$$

finally, we have the Calmar ratio as:

$$\text{Calmar} : \frac{\mathbb{E}[r]}{\text{MDD}} \quad (45)$$

4.2.3. Value at risk

It corresponds to the maximum expected loss with a probability α over a period.

$$\text{VaR}_\alpha(X) = \inf\{x : F_X(x) \geq \alpha\} = F_X^{-1}(\alpha) \quad (46)$$

4.2.4. Expected shortfall

The Expected Shortfall corresponds to the expected loss given that the loss exceeds the α -quantile of VaR (Value at Risk):

$$ES_\gamma(X) = \frac{1}{\gamma} \int_0^\gamma \text{VaR}_\alpha(X) d\alpha \quad (47)$$

Where $ES_\gamma(X)$ is the Expected Shortfall of the random variable X at a level γ , which means that given a loss r , we have $ES = E[r|r < \text{VaR}]$.

4.3. Model benchmarking

To compare the performance of the proposed framework, we introduce three base models that make a standard approach to the trading problem.

The selected models are equivalent to a policy to the DRL model, where the action is based as a function of the current market state representation S_t .

- **COIN:**

$$\pi_{COIN}(S_{t-1}) = \begin{cases} A_{t-1} = 1 & P_{t-1}^A \geq P_{t-1}^B \\ A_{t-1} = -1 & P_{t-1}^A < P_{t-1}^B \end{cases}$$

- **Support Vector Machine:** The proposed SVM model is made through a linear kernel to predict if the price difference of the synthetic assets ($P_t^S = P_t^A - P_t^B$) will go up or down:

$$\pi_{SVM}(S_{t-1}) = \begin{cases} A_{t-1} = 1 & \text{Up predicted} \\ A_{t-1} = -1 & \text{Down predicted} \end{cases}$$

- **Long Short-term Memory:** The proposed LSTM model is built as an LSTM layer with a Hyperbolic Tangent (tanh) activation of 64 neurons, a Dense layer of 32 neurons with a Rectified Linear Unit (ReLU) and a final Dense layer with two neurons with a Softmax activation output using the last three states to predict if the price difference of the synthetic assets ($P_t^S = P_t^A - P_t^B$) will go up or down:

$$\pi_{LSTM}(S_{t-3}, S_{t-2}, S_{t-1}) = \begin{cases} A_{t-1} = 1 & \text{Up predicted} \\ A_{t-1} = -1 & \text{Down predicted} \end{cases}$$

The training of the SVM and LSTM models differs from the DRL model; the optimization is based on a classification problem where the goal is to predict if the spread will increase or decrease, and the action is made as an independent problem of the overall training session. Meanwhile, the objective of the DRL is not to predict a 1-step action but to maximize the cumulative return over a trajectory of actions.

4.4. Daily performance

Generating the results for the different out-of-sample arbitrage strategies, we consider 150 trading days.

The strategies generated from the PPO, A2C, and DQN agents correspond to portfolio position directions that are changed or maintained at 30-minute intervals during a single day of trading. The COIN strategy represents the base arbitrage strategy from which improvement is expected through deep reinforcement learning. The strategy named Neutral corresponds to generating an arbitrage portfolio and maintaining a long and short position without changes for a day. This is a reference, considering no actions are taken to maximize returns. It can be observed that, compared to the other strategies, the neutral strategy, being a neutral position, maintains its purpose with minimal disturbance from market movements. This demonstrates that the generated portfolios can remain neutral if rebalanced daily over time. A summary of statistics for the log-returns of the testing period is presented in Table 2, and the results of the risk and performance metrics for the strategies are presented through Table 3.

Regarding risk measures, during the testing period, the base strategy COIN achieves the lowest risk as measured by Maximum Drawdown, Expected Shortfall at the 5% level, and standard deviation. The PPO agent exhibits the highest risk in three metrics, while the DQN agent ranks second highest in extreme losses measured by Expected Shortfall and Value at Risk. However, compared to the market, considering the best-performing asset during the period (LTC), the worst-performing asset (SOL), and a more established asset serving as the market benchmark (BTC), all strategies show a substantial improvement in terms of risk reduction. Although the strategies allocate half of the capital in long positions and the other half in short positions, while the market's "Buy & Hold" strategies have a long position with the entire capital, they are not directly comparable. Nonetheless, the strategies achieve much lower risk than the market. Considering the neutral strategy, similar risks are obtained compared to the active strategies, demonstrating the effectiveness of the portfolio construction strategy.

Regarding performance measures, when considering the risk-adjusted return based on different metrics, the DQN agent's strategy is the best during the period in terms of risk and return generated. With similar risks among the strategies, the cumulative return becomes the determining factor.

4.5. Intraday performance

Fig. 6 shows the Profit & Loss chart of intra-day strategies at a 30-minute frequency starting at 100%, including the neutral strategy, strategies implemented by DRL agents (PPO, DQN, and A2C), and the benchmark strategies COIN, SVM and LSTM. While the results are presented daily, they emerge from a trajectory of 47 decisions based on different criteria. Additionally, the intraday risks and the behavior

Table 2

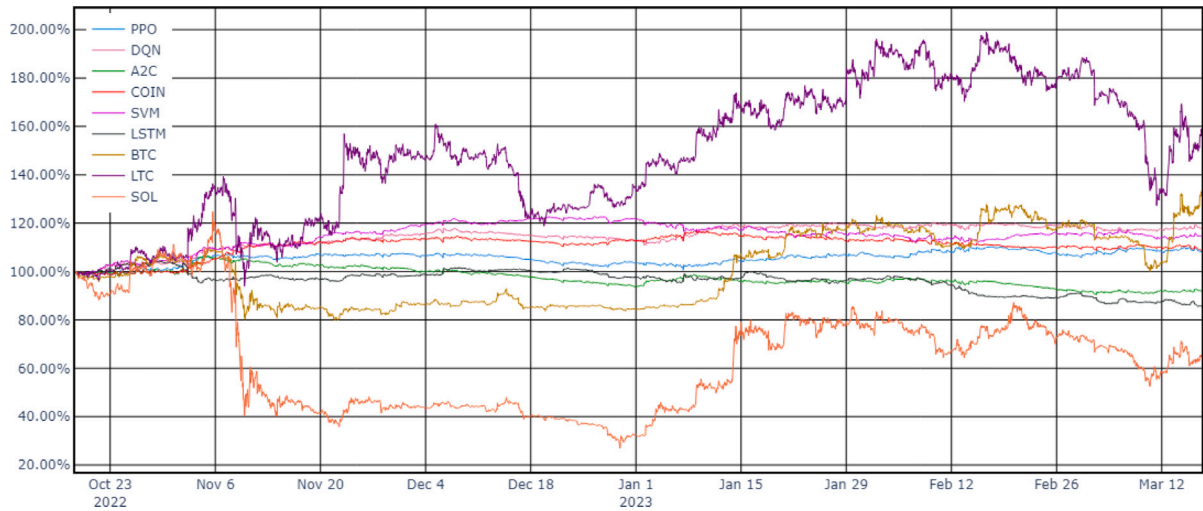
Summary of returns for different strategies.

	PPO	DQN	A2C	COIN	Neutral	SVM	LSTM	BTC	LTC	SOL
mean	0.46	0.51	0.39	0.48	0.01	0.41	0.14	0.19	0.30	-0.29
std	0.81	0.76	0.76	0.74	0.83	0.88	0.84	2.77	4.59	7.19
min	-2.38	-1.56	-1.01	-1.06	-2.43	-2.67	-3.46	-8.51	-11.92	-37.12
50%	0.36	0.47	0.33	0.43	0.02	0.47	0.21	0.08	0.12	-0.45
max	5.31	4.04	5.00	3.65	2.70	4.05	2.35	10.72	25.54	33.27
VaR 5%	-0.55	-0.69	-0.63	-0.66	-1.27	-0.97	-1.19	-3.95	-6.50	-9.94
ES 5%	-0.96	-1.04	-0.79	-0.79	-1.67	-1.44	-1.81	-5.79	-9.31	-16.32
skew	1.39	0.52	1.74	0.92	0.13	0.20	-0.48	0.66	1.09	-0.03
kurt	8.60	2.65	8.22	2.30	0.43	2.19	2.00	4.16	6.43	7.75

Table 3

Summary of risk and performance metrics for the different agents.

	MDD%	ES 5%	VaR 5%	$\sigma\%$	AS	Calmar	Sharpe	EPM	R%
PPO	2.35	-0.96	-0.56	0.81	0.0040	0.192	0.561	0.820	97.20
DQN	1.55	-1.04	-0.70	0.76	0.0039	0.325	0.664	1.220	112.82
A2C	1.08	-0.79	-0.64	0.76	0.0038	0.363	0.513	0.870	79.52
COIN	1.06	-0.79	-0.66	0.74	0.0039	0.454	0.648	1.102	105.10
SVM	2.64	-1.44	-0.97	0.88	0.0044	0.156	0.470	0.730	83.65
LSTM	7.11	-1.81	-1.19	0.84	0.0051	0.020	0.169	0.381	23.17
Neutral	3.22	-1.67	-1.27	0.83	0.0048	0.003	0.013	0.051	1.69
BTC	26.30	-5.79	-3.95	2.78	0.0156	0.007	0.068	0.048	32.51
LTC	34.82	-9.31	-6.50	4.59	0.0290	0.009	0.066	0.040	57.11
SOL	74.78	-16.32	-9.94	7.19	0.0363	-0.003	-0.040	-0.124	-35.70

**Fig. 6.** Profit & Loss chart of intra-day strategies at a 30-minute frequency.

of the strategies throughout their sequence of decisions are also studied to determine the efficiency of dollar-neutral positions and whether the agents manage to generate a significant difference in terms of intraday risk compared to the benchmark strategy COIN.

The results for intraday movements of the strategies exhibit the same trend as the aggregated results for a day. Firstly, all the means are positive, and even their medians (50th percentile) are positive, unlike the neutral strategy and the comparison assets (Table 4 summarizes statistics for the testing period intraday results). As expected, the minimum and maximum returns are less extreme than the reference assets.

On the other hand, when examining the strategies' correlations, it is observed that all strategies, including the neutral strategy, do not show a significant correlation with the reference assets. This reinforces that the portfolio construction is correct, and they have a beta very close

to 1, so a null beta is obtained by taking opposing positions (Table 5 presents the Pearson correlation matrix for the given strategies and benchmarks).

Table 6 shows a summary of risk and performance metrics for the different agents for intra-day returns. Strategies with lower risk based on the comparison metric are colored green, while those with higher risk are colored red. The three assets, BTC, LTC, and SOL, are also shown as market benchmarks for reference.

Compared to the daily results, regarding the selected risk metrics, the least risky agents are DQN and the benchmark strategy COIN, while the agents PPO and A2C turn out to be riskier. However, the results show that the risk is substantially reduced at the aggregated daily level and intraday level, around one-third of the risk of Bitcoin. Additionally, the efficiency of the arbitrage portfolio is maintained when controlling for the neutral strategy.

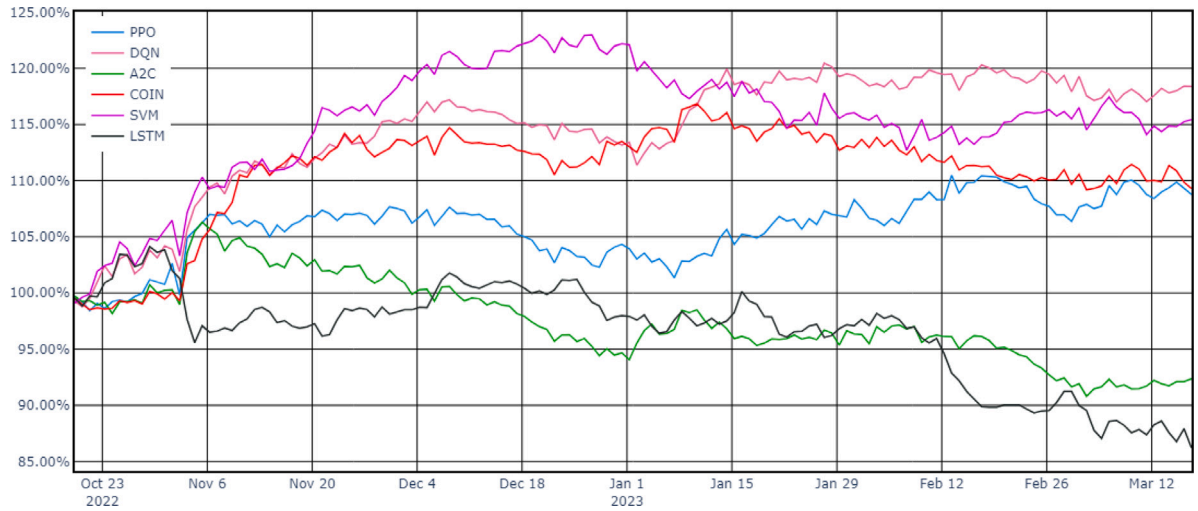


Fig. 7. Profit & Loss chart for daily results of strategies with a transaction cost of 0.02%.

Table 4

Summary statistics for intraday returns at a 30-minute frequency.

	PPO	DQN	A2C	COIN	Neutral	SVM	LSTM	BTC	LTC	SOL
mean	0.009	0.010	0.008	0.010	-0.000	0.008	0.003	0.004	0.006	-0.006
std	0.125	0.125	0.125	0.125	0.125	0.125	0.125	0.383	0.647	1.017
min	-1.152	-1.152	-1.152	-1.152	-1.344	-1.643	-2.571	-4.990	-9.811	-13.948
50%	0.002	0.002	0.000	0.002	0.000	0.004	0.002	-0.002	0.000	0.000
max	2.571	2.571	2.571	2.571	2.571	2.571	1.669	5.929	10.627	14.561
VaR 5%	-0.133	-0.129	-0.134	-0.132	-0.147	-0.134	-0.140	-0.450	-0.840	-1.211
ES5%	-0.217	-0.211	-0.217	-0.210	-0.266	-0.240	-0.271	-0.885	-1.447	-2.299
skew	3.950	4.054	4.158	4.195	1.776	2.700	-0.246	0.744	0.775	0.294
kurt	53.680	53.658	53.633	53.607	54.263	54.017	54.330	35.618	39.164	39.536

Table 5

Correlation matrix for the intra-day strategies and benchmarks.

	PPO	DQN	A2C	COIN	SVM	LSTM	BTC	LTC	SOL
PPO	1.00								
DQN	0.65	1.00							
A2C	0.65	0.71	1.00						
COIN	0.72	0.75	0.74	1.00					
SVM	0.48	0.55	0.46	0.49	1.00				
LSTM	0.02	0.09	0.03	0.04	0.10	1.00			
BTC	0.01	0.00	0.01	-0.00	-0.00	-0.00	1.00		
LTC	0.05	0.04	0.05	0.04	0.04	-0.04	0.67	1.00	
SOL	0.01	0.00	0.02	-0.01	0.00	0.01	0.70	0.59	1.00

Regarding their performance metrics, the DQN and COIN strategies continue to have the best performance, with DQN outperforming in the Sharpe ratio and COIN in Calmar and EPM. In this case, the A2C agent performs the worst in all metrics.

4.6. Transaction costs

Measuring the performance of strategies under transaction costs, considering that the strategy applies for one day, possibly restructuring the position multiple times during the day, with a maximum of 47 times daily. The profit and loss for daily strategies results when applying a transaction cost of 0.02% per trade are presented in Fig. 7. The Benchmark corresponds to the base COIN strategy. Although the agents were not trained in considering transaction costs, the impact of these costs on high-frequency strategies is studied. Table 7 summary of statistics for the results when considering transaction costs. The mean, 50%, and VaR are lower than the no transaction costs case.

Compared to the strategies without considering transaction costs, this time, the agent trained by DQN outperforms the benchmark strategy COIN in all performance metrics (Table 8 summarizes the selected

risk and performance metrics when considering transaction costs of 0.02%). Regarding risk measures, the DQN agent performs the best in three risk metrics and the worst in one. Concerning tail risks or maximum losses generated, the VaR and ES metrics show that DQN exposes itself to higher losses than the other strategies. However, regarding VaR, the differences among all strategies are minimal. Regarding unconditional volatility, all strategies have practically identical results except for PPO, which is 0.05% more volatile. As for the maximum drawdown, the DQN strategy has the lowest accumulated maximum loss in the sample, while A2C has the highest loss. This measure shows that although they all have similar levels of volatility and tail risks, there are differences in their management and ability to generate returns by controlling losses beyond a particular trade.

Regarding the AS risk measure that considers different moments of the unconditional distribution of returns, it is found that DQN remains the least risky strategy. Considering that all strategies have similar volatility and extreme losses, this measure confirms what is observed in the maximum drawdown.

4.7. Agent activity

Although the ability of the different strategies to generate returns and the level of risk they expose, along with their risk-adjusted returns, were measured, the agent's ability to predict market movements beyond whether the return is positive or negative was not assessed.

For the overall activity level, the Unconditional Coverage (UC) test by Kupiec (1995) [28] is conducted to test whether the actions are random under the null hypothesis that the actions are random with equal probability:

Under the assumption of a correct model specification, the sequence of actions should be independent and follow a Bernoulli distribution.

$$H_0 : A_t \sim \text{Bernoulli}(\alpha), \quad (48)$$

Table 6
Summary of risk and performance metrics for the different agents for intraday returns.

	MDD	ES 5	VaR 5	σ	AS	Calmar	Sharpe	ESM
PPO	2.526	-0.217	-0.133	0.1250	0.00092	0.0037	0.0747	0.01646
DQN	2.235	-0.211	-0.129	0.1249	0.00092	0.0047	0.0833	0.02049
A2C	2.421	-0.217	-0.134	0.1251	0.00091	0.0033	0.0644	0.00306
COIN	1.983	-0.210	-0.132	0.1250	0.00091	0.0050	0.0792	0.02547
SVM	3.078	-0.240	-0.134	0.1251	0.00093	0.0032	0.0667	0.00311
LSTM	7.625	-0.271	-0.140	0.1253	0.00096	0.0004	0.0225	0.00323
Neutral	4.035	-0.266	-0.147	0.1254	0.00091	0.0000	-0.0013	0.00000
BTC	27.023	-0.885	-0.450	0.383	0.00215	0.0001	0.0104	-0.00778
LTC	36.117	-1.447	-0.840	0.647	0.02139	0.0002	0.0098	0.00303
SOL	78.471	-2.299	-1.211	1.0169	0.08455	-0.0001	-0.0058	-0.00065

Table 7
Summary statistics for daily strategy considering transaction costs.

	PPO [%]	DQN [%]	A2C [%]	COIN [%]	SVM [%]	LSTM [%]
mean	0.061	0.118	-0.051	0.063	0.102	-0.097
std	0.822	0.772	0.766	0.756	0.886	0.838
min	-2.789	-1.942	-1.462	-1.495	-3.020	-3.667
50%	-0.057	0.108	-0.089	0.016	0.176	-0.035
max	4.998	3.617	4.549	3.251	3.667	2.101
VaR 5%	-0.996	-1.093	-1.083	-1.108	-1.268	-1.408
ES5%	-1.395	-1.448	-1.235	-1.224	-1.786	-2.050
skew	1.383	0.466	1.713	0.892	0.121	-0.459
kurt	8.494	2.269	7.925	2.125	1.982	1.884

$$f(A_t, p) = (1 - p)^{1-A_t} p^{A_t} \quad (49)$$

For the actions $\{A_t\}_t$ where $A_t = 0$ is the action of maintaining the current position and $A_t = 1$ correspond to change direction,

$$L(\pi) = \prod_t (1 - \pi)^{1-A_t} \pi^{A_t} = (1 - \pi)^{T_0} \pi^{T_1} \quad (50)$$

then the maximum likelihood estimator (MLE),

$$\hat{\pi}_{MLE} = \frac{T_1}{T_0 + T_1} \quad (51)$$

T_0 is the total amount of $A_t = 0$ and T_1 to the amount of $A_t = 1$. Under the null hypothesis $H_0 : \pi = \alpha$

$$LR_{uc} = -2 \ln \left[\frac{L(\alpha)}{L(\hat{\pi})} \right] \sim \chi_1^2, \quad (52)$$

Therefore, the objective is to reject the null hypothesis and conclude that the agent does not act randomly.

Additionally, considering the conditional distribution of actions, we have the Conditional Coverage (CC) test by Christoffersen (1998) [29].

Considering that $\{A_t\}_t$ exhibits temporal dependence and follows a Markov sequence with a transition probability matrix:

$$\Pi = \begin{pmatrix} 1 - \pi_{01} & \pi_{01} \\ 1 - \pi_{11} & \pi_{11} \end{pmatrix}, \quad (53)$$

where π_{ij} with $i, j \in 0, 1$ is the transition probability from state i to state j with $T_{i,j}$ the total amount of actions transitioning from state i to state j , the likelihood function is given by:

$$L(\Pi) = (1 - \pi_{01})^{T_{00}} \pi_{01}^{T_{01}} (1 - \pi_{11})^{T_{10}} \pi_{11}^{T_{11}}, \quad (54)$$

If $\{A_t\}_t$ is independent of time, then $\pi_{01} = \pi_{11} = \pi$, and we have:

$$\hat{\Pi}_{MLE} = \begin{pmatrix} 1 - \hat{\pi} & \hat{\pi} \\ 1 - \hat{\pi} & \hat{\pi} \end{pmatrix}, \quad LR_{cc} = -2 \ln \left[\frac{L(\alpha)}{L(\hat{\Pi})} \right] \sim \chi_1^2. \quad (55)$$

Therefore, we aim to reject the null hypothesis that the actions are random and independent of each other.

Test results for the tests and MLE are presented in Table 9. The MLE was used to estimate the transition and unconditional probabilities relative to the total actions taken in each state. These actions were derived from 47 daily trades for 150 days, resulting in 7,050 actions

executed. The results of the conducted test are reported based on their respective p -values to reject the null hypothesis at a significance level of 5%. Thus, we can reject the assumption that the actions taken by the different agents are random with equal probabilities.

4.8. Prediction ability

In terms of the predictive ability regarding the price movement direction of the synthetic asset, we examine whether the actions predict such movement. The standard metric of classification problems are based on the following values:

- **True Positive:** Price difference will go up and go up.
- **True Negative:** Price difference will go down.
- **False Positive:** Price difference will go up and down.
- **False Negative:** Price difference will go down and up.

It is important to consider the meaning of each metric for the purpose of the studied problem. The return on the prediction will also depend on the previous state, considering the turnover cost of the current position. Each time the direction of the position is changed, there is a cost of closing the current position and opening it in the inverse direction so that a constant position change will harm the result independent of the prediction result.

TP and TN will yield a return over market movement; meanwhile, FP and FN will yield a loss due to market movement. This metric does not account for transaction costs.

Considering this, the precision score does not have any useful meaning because TP and TN end in a positive return while FN and FP end in a negative return, so the results are only two rather than 4.

The standard prediction metrics are 4, Accuracy, Precision, Recall, and F1 Score:

• Accuracy:

It corresponds to the total number of correct predictions divided by the total number of cases.

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + FN + TN} \quad (56)$$

• Precision:

It corresponds to the proportion of positive predictions that are positive.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (57)$$

• Recall:

It corresponds to the true positives divided by the total number of positive cases.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (58)$$

• F1 Score:

It corresponds to the true positives divided by the total number of positive cases.

$$\text{F1 Score} = \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (59)$$

Table 8

Summary of risk and performance metrics for comparing strategies when transaction costs are taken.

	MMD[%]	ES 5%	VaR 5%	σ [%]	AS	Calmar	Sharpe	EPM	R[%]
PPO	5.87	-1.39	-0.99	0.82	0.0036	0.0103	0.0738	0.1558	8.77
DQN	4.96	-1.44	-1.09	0.77	0.0030	0.0239	0.1533	0.4004	18.39
A2C	14.57	-1.23	-1.08	0.77	0.0032	-0.0035	-0.0670	-0.1430	-7.63
COIN	6.56	-1.22	-1.10	0.76	0.0037	0.0096	0.0834	0.1597	9.29
SVM	8.36	-1.79	-1.27	0.89	0.0038	0.0122	0.1156	0.1521	15.43
LSTM	17.21	-2.05	-1.41	0.84	0.0041	-0.0056	-0.1153	-0.1549	-13.79

Table 9

The Maximum Likelihood Estimation (MLE) and test results.

Agent	State 1 ($\hat{x}$)	State 0	$\hat{x}_{0,0}$	$\hat{x}_{0,1}$	$\hat{x}_{1,0}$	$\hat{x}_{1,1}$	UC	CC
PPO	42.44 [%]	57.56 [%]	55.07 [%]	44.93 [%]	60.92 [%]	39.07 [%]	<1e6	<1e6
DQN	47.79 [%]	52.21 [%]	50.57 [%]	49.43 [%]	54.02 [%]	45.98 [%]	<1e6	<1e6
A2C	70.27 [%]	29.73 [%]	31.74 [%]	68.26 [%]	28.89 [%]	71.11 [%]	<1e6	<1e6
COIN	48.31 [%]	51.69 [%]	48.09 [%]	51.91 [%]	55.52 [%]	44.48 [%]	<1e6	<1e6

Table 10

Results of spread prediction for all the compared models.

Agent	Accuracy	Precision	Recall	F1
PPO	51.64%	53.81%	44.28%	48.58%
DQN	52.01%	53.79%	49.69%	51.66%
A2C	51.35%	52.10%	70.74%	60.01%
COIN	52.10%	53.83%	50.36%	52.04%
SVM	53.42%	54.73%	56.23%	55.47%
LSTM	52.01%	53.55%	52.73%	53.14%

Precision, Recall, and F1 Score are not complete metrics in this approach because they take only a fraction of the results. Accuracy is the only metric that has meaning because it compares the proportion of positive returns, which is the ratio of actions that made a return from market movement overall actions. It is important that this does not consider cases where the action return is less than the cost arising from the position turnover.

The resulting metrics are not too different; for example, DQN, COIN, and LSTM have practically the same scores but have very different results (Table 10). Regarding risk, DRL agents are less risky than the SVM and LSTM models; meanwhile, regarding prediction accuracy, the SVM model outperforms all DRL models, which is expected because SVM is optimized on pure prediction rather than a cumulative return of the actions.

5. Conclusions and further research

This paper presents a deep reinforcement learning-based trading method for generating profits in the cryptocurrency market. The method utilizes a unified framework for arbitrage portfolio generation, market signal decomposition and extraction, and decision-making to select positions on synthetic assets. The proposed method uses deep reinforcement learning with three different types of agents (algorithms) for decision-making: PPO, DQN, and A2C.

Portfolios are generated based on the cointegration relationship of different assets. Market signals are extracted using various technical indicators and transformations, defining the DRL scheme's market states. Finally, a market scenario simulator is created using a large historical window for network training. A 7-day window is used for portfolio generation, and a 1-day window (47 decisions) is used for the agent to take actions to generate profits.

An extensive out-of-sample analysis is conducted at the daily level considering the aggregated result of the agent's decision trajectory throughout a one-day session and at the intra-day level considering each individual decision. The analysis focuses on the risk of the generated strategies using various risk measures and also studies the

risk-adjusted performance using different metrics. The analysis is conducted both with and without transaction costs, along with a study of the actions taken by the agents to determine if there is coherent behavior rather than pure randomness.

According to the tests conducted, the proposed method is capable of generating profits and reducing risk in the highly volatile cryptocurrency market and remains profitable even when transaction costs are considered. Additionally, the agent's activity results show that the generated actions are not random and provide a substantial improvement (particularly with DQN) over the base arbitrage strategy (COIN).

Compared with other standard models like SVM and LSTM as a pure prediction model that is an in-dependent action decision-making model, SVM still achieves a better accuracy score, but the DQN model yields a better total return, less risk, and a better risk-adjusted return.

While the results of the proposed methodology demonstrate profitability under transaction costs, the paper does not delve into the utilized network's specific structure or consider a more granular action space.

CRedit authorship contribution statement

Gabriel Vergara: Conceptualization, Data curation, Formal analysis, Methodology, Writing – original draft. **Werner Kristjanpoller:** Conceptualization, Formal analysis, Methodology, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] E. Gatev, W.N. Goetzmann, K.G. Rouwenhorst, Pairs trading: Performance of a relative-value arbitrage rule, *Rev. Financ. Stud.* 19 (3) (2006) 797–827.
- [2] P. Wolf, C. Hubschneider, M. Weber, A. Bauer, J. Härtl, F. Dürr, J. Zöllner, Learning how to drive in a real world simulation with deep q-networks, in: 2017 IEEE Intelligent Vehicles Symposium, IV, IEEE, 2017, pp. 244–250.
- [3] H. Beom, H. Cho, A sensor-based navigation for a mobile robot using fuzzy logic and reinforcement learning, *IEEE Trans. Syst., Man, Cybern.* 25 (3) (1995) 464–477.
- [4] D. Silver, T. Hubert, J. Schrittwieser, I. Antonoglou, M. Lai, A. Guez, M. Lanctot, L. Sifre, D. Kumaran, T. Graepel, et al., A general reinforcement learning algorithm that masters chess, shogi, and go through self-play, *Science* 362 (6419) (2018) 1140–1144.
- [5] R. Neuneier, Enhancing Q-learning for optimal asset allocation, in: *Advances in Neural Information Processing Systems*, vol. 10, 1997.
- [6] Y. Li, Deep reinforcement learning: An overview, 2017, arXiv preprint [arXiv:1701.07274](https://arxiv.org/abs/1701.07274).
- [7] Y. Deng, F. Bao, Y. Kong, Z. Ren, Q. Dai, Deep direct reinforcement learning for financial signal representation and trading, *IEEE Trans. Neural Networks Learn. Syst.* 28 (3) (2016) 653–664.
- [8] Z. Jiang, D. Xu, J. Liang, A deep reinforcement learning framework for the financial portfolio management problem, 2017, arXiv preprint [arXiv:1706.10059](https://arxiv.org/abs/1706.10059).
- [9] J. Carapuço, R. Neves, N. Horta, Reinforcement learning applied to Forex trading, *Appl. Soft Comput.* 73 (2018) 783–794.
- [10] Z. Liang, H. Chen, J. Zhu, K. Jiang, Y. Li, Adversarial deep reinforcement learning in portfolio management, 2018, arXiv preprint [arXiv:1808.09940](https://arxiv.org/abs/1808.09940).
- [11] J. Lee, R. Kim, S. Yi, J. Kang, MAPS: Multi-agent reinforcement learning-based portfolio management system, 2020, arXiv preprint [arXiv:2007.05402](https://arxiv.org/abs/2007.05402).
- [12] X. Wu, H. Chen, J. Wang, L. Troiano, V. Loia, H. Fujita, Adaptive stock trading strategies with deep reinforcement learning methods, *Inform. Sci.* 538 (2020) 142–158.
- [13] Z. Zhang, S. Zohren, S. Roberts, Deep reinforcement learning for trading, *J. Financ. Data Sci.* 2 (2) (2020) 25–40.
- [14] F. Liu, Y. Li, B. Li, J. Li, H. Xie, Bitcoin transaction strategy construction based on deep reinforcement learning, *Appl. Soft Comput.* 113 (2021) 107952.
- [15] Y. Chen, S. Huang, Sentiment-influenced trading system based on multimodal deep reinforcement learning, *Appl. Soft Comput.* 112 (2021) 107788.
- [16] Y. Lin, C. Chen, C. Sang, S. Huang, Multiagent-based deep reinforcement learning for risk-shifting portfolio management, *Appl. Soft Comput.* 123 (2022) 108894.
- [17] J. Mulvey, Y. Sun, M. Wang, J. Ye, Optimizing a portfolio of mean-reverting assets with transaction costs via a feedforward neural network, *Quant. Finance* 20 (8) (2020) 1239–1261.
- [18] T. Kim, H. Kim, Optimizing the pairs-trading strategy using deep reinforcement learning with trading and stop-loss boundaries, *Complexity* 2019 (2019) 1–20.
- [19] J. Guíjarro-Ordóñez, M. Pelger, G. Zanotti, Deep learning statistical arbitrage, 2021, arXiv preprint [arXiv:2106.04028](https://arxiv.org/abs/2106.04028).
- [20] J. Durbin, S. Koopman, *Time Series Analysis By State Space Methods*, Vol. 38, OUP Oxford, 2012.
- [21] V. Mnih, K. Kavukcuoglu, D. Silver, A. Graves, I. Antonoglou, D. Wierstra, M. Riedmiller, Playing atari with deep reinforcement learning, 2013, arXiv preprint [arXiv:1312.5602](https://arxiv.org/abs/1312.5602).
- [22] V. Mnih, K. Kavukcuoglu, D. Silver, A. Rusu, J. Veness, M. Bellemare, A. Graves, M. Riedmiller, A. Fidjeland, G. Ostrovski, et al., Human-level control through deep reinforcement learning, *Nature* 518 (7540) (2015) 529–533.
- [23] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, O. Klimov, Proximal policy optimization algorithms, 2017, arXiv preprint [arXiv:1707.06347](https://arxiv.org/abs/1707.06347).
- [24] V. Mnih, A. Badia, M. Mirza, A. Graves, T. Lillicrap, T. Harley, D. Silver, K. Kavukcuoglu, Asynchronous methods for deep reinforcement learning, in: *International Conference on Machine Learning*, PMLR, 2016, pp. 1928–1937.
- [25] R. Aumann, R. Serrano, An economic index of riskiness, *J. Polit. Econ.* 116 (5) (2008) 810–836.
- [26] U. Himm, C. Pigorsch, Beyond the Sharpe ratio: An application of the Aumann–Serrano index to performance measurement, *J. Bank. Financ.* 36 (2012) 2274–2284.
- [27] T. Li, Y. Kim, Q. Fan, F. Zhu, Aumann–Serrano index of risk in portfolio optimization, *Math. Methods Oper. Res.* 94 (2) (2021) 197–217.
- [28] P. Kupiec, et al., Techniques for Verifying the Accuracy of Risk Measurement Models, Vol. 95, no. 24, Division of Research and Statistics, Division of Monetary Affairs, Federal ..., 1995.
- [29] P. Christoffersen, Evaluating interval forecasts, *Int. Econ. Rev.* (1998) 841–862.